

The technical report, judged against the ideal functionality

For Halil, 2026-08-29

The frame

You said you value that the work is discussed along the ideal functionality. That gives a clean test of completeness, because a work built on $\mathcal{F}$ is complete when three things hold and not before.

1. $\mathcal{F}$ says what we mean. The functionality is the definition of the goal, so a defect in it is a defect in everything downstream.
2. The protocol realizes $\mathcal{F}$. Every theorem is proved and every hypothesis is one we meet.
3. What $\mathcal{F}$ itself leaks is measured. Realizing an ideal functionality says nothing about what that functionality reveals, and that gap is where the whole privacy story lives.

Below is what is missing at each layer. Nothing here is prose. It is defects, proofs and runs.

Layer 1. The functionality does not say what we mean

Step 3 does not branch on the selection. Functionality 1 step 3 returns $\arg \max_c (\theta^* \varphi_j(x))_c$ whatever a^* was. The map φ_j is the backbone composed with client j 's adapter, so the functionality always serves the personal-adapter arrangement and never models the bare-backbone one. Step 2 announces a choice that step 3 ignores.

This is the central formal object and it is wrong. One clause fixes it. It is first on the list because Theorem 1 proves the protocol realizes this functionality, so the theorem currently proves the wrong statement.

The allowance is a counter with no adversary model. $\mathcal{F}$ meters per client. A coalition of $t - 1$ clients therefore commands $(t - 1)Q$, which Theorem 2 uses and $\mathcal{F}$ never states. Worth one sentence so the two agree.

Layer 2. The proofs, and the one hypothesis we do not meet

Theorem 1's simulation needs a smudging condition our parameters miss. The key-switch simulator we invoke is correct when the flooding dominates the circuit noise. Ours is about 2^{20} the wrong way. Section IV-G states this. It is a gap in provenance rather than a break, and the honest options are to derive a parameter for our circuit, or to keep the statement as it is and let the condition be an assumption. Cheap either way, and it is the thing a cryptography reviewer checks first.

Proposition 2 caps δ and nobody has measured the cap. This is the largest hole in the paper. The proposition says the coalition's membership advantage is bounded by what the head

itself reveals, at every budget. That turns a vacuous bound into a real one only once δ_{wb} has a number. It needs the mia/ rewire and one VALAR session.

Proposition 1’s hypothesis is not checked against the protocol. It proves impossibility for a predicate whose view is independent of the plaintext, and we apply it to a client whose view includes its own data. Either check the hypothesis or narrow the conclusion.

Layer 3. What is measured, and the four holes

Measured and recorded: extraction cost and its Cd law, the noise defence, accuracy on five tasks, CIFAR-10 on two published partitions, communication against the client count, the two level-restoration mechanisms, and the GPU restoration.

Membership inference. No record. The mia/ suite attacks a released model and a prototype channel, neither of which exists. Proposition 2 makes this a white-box attack on the true merged head rather than an extract-then-attack pipeline, so it is smaller than it was, but it is still a rewire and a run.

The selection phase has no cost. The submission now says at most $2NC$ encrypted comparisons and that bound is arithmetic, not a measurement. The circuit is one encrypted comparison per held-out example per arrangement. Worth one job.

The argmax index under the specified restoration. This is nearly closed. The index is measured and exact, and `tracked_index` costs 4 to 9 per cent over the maximum. But it was measured under collective refresh, and we now report server-side bootstrapping, so the number we print is for a circuit that returns the maximum. One job closes it.

One real end-to-end query. Every serving benchmark runs on synthetic feature vectors. A Python exporter and about a hundred lines of Go.

Free, and worth doing first

Fold the twelve CIFAR-10 cells into the selection table. It turns 13 of 15 into 24 of 27 and costs nothing.

Delete the gradient-reconstruction sentence from related work. You deleted it from the introduction because a USENIX Security 2025 SoK contradicts it for a semi-honest server, and it survives verbatim in Section VI.

The order I would take it

Free first, then the formal defects, then the jobs, because a job that measures a protocol whose functionality is wrong measures the wrong thing.

1. The two free items above.
2. Functionality step 3 branches on a^* . One clause, and it is the object every theorem points at.
3. Decide the smudging statement. Assumption or derivation.
4. Submit the three jobs, which are independent and can run together. The argmax index under server-side bootstrapping. The selection cost. The membership measurement, after the mia/ rewire.

5. Write the report's missing sections from what lands.

Items 1 to 3 are hours and need no compute. Item 4 is where the calendar goes.